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Explainable machine learning for long-term cardiovascular disease risk prediction in Chinese middle-aged and older adults: a 9-year longitudinal cohort study with web-based risk calculator

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Cardiovascular disease represents the leading cause of mortality in China, accounting for over 40% of all deaths. Existing risk prediction models predominantly derive from Western populations, rendering them suboptimally calibrated for the Chinese middle-aged and elderly demographic. Conventional statistical approaches inadequately capture non-linear associations within high-dimensional data, whilst machine learning models, despite superior performance, suffer from insufficient interpretability. This study leveraged a nationally representative cohort to develop an interpretable machine learning-based tool for long-term cardiovascular risk prediction tailored to the Chinese population. To compare the predictive performance of ten machine learning algorithms using data from the China Health and Retirement Longitudinal Study (CHARLS), identify the optimal model, achieve transparent interpretation through SHapley Additive exPlanations (SHAP) methodology, and develop an individualized cardiovascular risk assessment tool for Chinese residents aged 45 years and above. The study enrolled 8,080 participants aged ≥ 45 years without baseline cardiovascular disease from the CHARLS 2011–2020 longitudinal dataset, with 9-year follow-up. The primary outcome was incident cardiovascular disease. From 77 candidate variables, logistic regression analysis identified 11 predictors: geographical region, hypertension, dyslipidaemia, liver disease, asthma, depression score, age, sleep duration, triglycerides, high-density lipoprotein cholesterol, and waist circumference. The cohort was randomly partitioned into training (n = 5,657, 70%) and validation (n = 2,423, 30%) sets. Ten predictive models were constructed, including random forest, gradient boosting machine, and extreme gradient boosting. Model performance was evaluated using area under the receiver operating characteristic curve (AUROC), calibration plots, and decision curve analysis. Feature contributions were elucidated using SHAP values. Incident cardiovascular disease occurred in 1,246 participants (22.0%) within the training cohort. Multivariable analysis identified hypertension (adjusted OR 1.80), waist circumference (adjusted OR 1.05 per 1-cm increment), dyslipidaemia (adjusted OR 1.42), and liver disease (adjusted OR 1.60) as principal independent predictors. Among ten algorithms evaluated, random forest demonstrated superior performance: validation set AUC 0.829 (95% CI 0.809–0.848), accuracy 0.770, sensitivity 0.681, specificity 0.795. The model exhibited excellent calibration and yielded maximal net clinical benefit across the 10%-85% risk threshold spectrum. SHAP analysis revealed waist circumference as the predominant contributor, followed by triglycerides, age, and hypertension. Psychobehavioural factors (depression, sleep duration) demonstrated independent predictive value. A web-based risk calculator was developed, providing real-time individual 9-year cardiovascular disease probability estimates. The random forest model accurately predicts cardiovascular disease risk in the Chinese middle-aged and elderly population, with waist circumference emerging as the most critical predictor. Translated into an online assessment tool, this

model facilitates community-based screening and individualized prevention, offering a pragmatic risk stratification approach for resource-constrained settings.

Keywords Cardiovascular disease, Risk prediction, Machine learning, Random forest, SHAP (SHapley Additive exPlanations), Abdominal obesity, Waist circumference, Longitudinal cohort, China Health and Retirement Longitudinal Study (CHARLS), Explainable artificial intelligence, Web-based calculator

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Cardiovascular disease (CVD) has emerged as the foremost cause of mortality worldwide. The Global Burden of Disease Study 2021 documented approximately 20.5 million CVD-attributable deaths, representing 32.4% of global mortality, with ischaemic heart disease and stroke accounting for 9.2 million and 7.6 million deaths respectively, whilst contributing 430 million disability-adjusted life-years lost[1]. This public health challenge assumes particular gravity in China. The Report on Cardiovascular Health and Diseases in China 2024 estimates the current prevalence exceeds 350 million cases, encompassing 14 million stroke survivors, 12 million individuals with coronary heart disease, 9 million with heart failure, and a striking 245 million with hypertension[2]. In 2022, the CVD mortality rate reached 324.49 per 100,000, constituting the leading cause of death amongst both rural (48.12%) and urban (46.74%) residents[2]. Despite declining age-standardised mortality rates, accelerated population ageing continues to amplify the absolute disease burden—by 2020, adults aged ≥ 60 years numbered 264 million (18.70% of the population), with projections estimating 40 million CVD patients by 2030[3]. Within this epidemiological context, accurate identification of high-risk individuals and implementation of early intervention strategies constitute pivotal imperatives for disease control.

Risk prediction models represent the cornerstone of precision CVD prevention. Since the seminal 1998 Framingham Heart Study risk score[4], such tools have gained widespread adoption in clinical decision-making. The American Heart Association/American College of Cardiology Pooled Cohort Equations (PCE)[5], the European Systematic Coronary Risk Evaluation (SCORE) system[6], and the WHO cardiovascular risk charts have been extensively deployed across Western populations. However, the cross-population generalisability of these models exhibits substantial limitations. A validation study encompassing 95,649 Chinese adults demonstrated systematic risk overestimation and suboptimal calibration of the Framingham score in Chinese cohorts[7]. External validation of PCE equations in Chinese populations yielded similarly disappointing results[8], with predictive discordance attributable to ethnic genetic heterogeneity, lifestyle patterns, risk factor prevalence profiles, and healthcare system disparities[9].

In response to the inadequate localisation of international tools, Chinese investigators have undertaken systematic endeavours. Leveraging data from 127,000 participants in the China Prediction for Atherosclerotic Cardiovascular Disease Risk (China-PAR) study, researchers developed a 10-year atherosclerotic CVD risk prediction instrument for adults aged 35–79 years, achieving an external validation C-statistic of 0.794[10]. The Kailuan cohort and China Patient-centered Evaluative Assessment of Cardiac Events (PEACE) Million Persons Project have subsequently established indigenous prediction frameworks[11–13]. Nevertheless, these models predominantly employ Cox proportional hazards or logistic regression, with performance constrained by statistical assumptions. Conventional methodologies inadequately capture non-linear relationships and interaction effects amongst high-dimensional variables[14], whilst demonstrating limitations in handling missing data and multicollinearity[15,16]. A systematic review incorporating 32 traditional CVD prediction models revealed a median external validation C-statistic of merely 0.74 (IQR 0.70–0.78), with uniformly suboptimal calibration[17].

Machine learning algorithms demonstrate superiority in addressing complex non-linear associations, yet their "black-box" opacity impedes clinical translation. Ensemble methods such as random forest and gradient boosting machines, despite exhibiting exceptional predictive performance, possess opaque internal decision architectures that provoke trust and ethical concerns in high-stakes medical scenarios[18,19]. Explainable artificial intelligence (XAI) technologies offer a resolution to this dilemma. SHapley Additive exPlanations (SHAP), grounded in game-theoretic Shapley value principles, provides unified feature contribution quantification for arbitrarily complex models[20,21]. Compared with traditional feature importance metrics, SHAP satisfies mathematical properties including additivity, symmetry, dummy variable consistency, and efficiency, whilst facilitating both global feature ranking and individualised patient-level explanations[22]. This methodology has been validated across oncological prognostication[23], diabetic complication assessment[24], and intensive care unit mortality prediction[25]. Nevertheless, systematic application of interpretable machine learning to long-term CVD risk prediction in China's middle-aged and elderly populations remains conspicuously absent.

The China Health and Retirement Longitudinal Study (CHARLS) furnishes an optimal platform to address this lacuna. Initiated in 2011, the programme encompasses 28 provinces, 150 counties, and 450 communities, employing multistage probability sampling to recruit approximately 23,000 residents aged ≥ 45 years, with follow-up assessments conducted biennially to triennially[26]. The survey architecture incorporates multidimensional data spanning demographic characteristics, socioeconomic indicators, functional health status, lifestyle behaviours, anthropometric measurements, and biomarkers[27]. Its nationally representative sampling framework, variable breadth, and longitudinal design provide a robust foundation for comprehensive prediction model development.

The present investigation leverages CHARLS 2011–2020 longitudinal data to systematically compare ten machine learning algorithms for long-term CVD risk prediction in China's middle-aged and elderly demographic, employs SHAP methodology to elucidate model decision mechanisms and critical risk determinants, and translates the optimal model into an online individualised risk assessment platform. This study integrates

conventional risk factors with psychosocial determinants, ensures model generalisability through nationally representative cohort derivation, enhances clinical utility via interpretability analysis, and ultimately transforms research outputs into pragmatic tools, thereby furnishing scientific evidence and decision support for precision CVD prevention and control amongst China's ageing population.

Methods

Study design and data source

This population-based prospective cohort study utilised data from the China Health and Retirement Longitudinal Study (CHARLS)[26]. Administered by the National School of Development at Peking University, CHARLS employed multistage stratified probability-proportional-to-size (PPS) sampling to recruit approximately 17,000 community-dwelling residents aged ≥ 45 years and their spouses from 450 village or urban committees across 150 counties (districts) in 28 provinces, autonomous regions, and municipalities.

Baseline assessment was conducted in 2011, with subsequent follow-up waves completed in 2013, 2015, 2018, and 2020, constituting a 9-year observation period. This investigation integrated longitudinal data from all five temporal nodes for cardiovascular disease long-term risk prediction model development and validation. The study protocol received approval from the Biomedical Ethics Review Committee of Peking University (approval number: IRB00001052-11015), with written informed consent obtained from all participants.

Participant Inclusion and Exclusion Criteria

The baseline survey enrolled 17,708 participants. Exclusion criteria comprised: (1) physician-diagnosed myocardial infarction, coronary heart disease, angina pectoris, congestive heart failure, or stroke at baseline; (2) indeterminate baseline cardiovascular disease history; (3) absence of blood biochemistry assays; (4) > 20% missingness in core predictor variables; (5) loss to follow-up or incomplete outcome ascertainment during the observation period. Given systematic deficiencies in mortality documentation during the 2015 and 2018 survey waves, effective sample sizes for corresponding periods underwent calibration. Ultimately, 8,080 participants were enrolled in the model development cohort, with detailed selection workflow presented in Fig. 1.

Outcome definition and adjudication

Incident CVD served as the primary outcome, encompassing first diagnosis of cardiac events (coronary heart disease, angina pectoris, myocardial infarction, congestive heart failure, or other cardiac conditions) or stroke. This composite definition reflects our focus on overall atherosclerotic cardiovascular burden rather than individual disease entities. We recognize that grouping pathophysiologically diverse conditions—ischemic heart disease, heart failure with potentially disparate origins, and cerebrovascular events—may dilute subtype-specific risk associations. Nevertheless, this approach aligns with primary prevention goals: identifying individuals requiring global cardiovascular risk modification rather than targeting specific disease trajectories. Outcome ascertainment relied on participant self-report of physician diagnosis via standardised questionnaires, a methodology validated in prior CHARLS cohort investigations[28,29]. CVD events were adjudicated exclusively

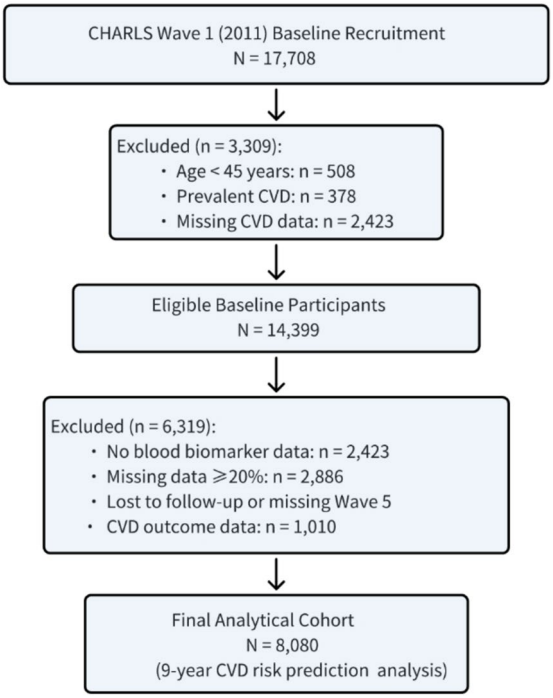


Fig. 1. Study participant selection flowchart.

when participants without baseline disease history reported new diagnoses at subsequent follow-up assessments, with event timing defined as the corresponding visit date.

Candidate predictor variables and selection strategy

Seventy-seven baseline candidate variables were initially incorporated, encompassing demographic characteristics (age, sex, geographical region, marital status, educational attainment), medical history (hypertension, dyslipidaemia, diabetes mellitus, hepatic disease, asthma), lifestyle behaviours (smoking, alcohol consumption, physical activity, sleep duration), anthropometric measurements (height, weight, waist circumference, body mass index), psychological assessment (10-item Centre for Epidemiologic Studies Depression Scale [CESD-10]), and laboratory parameters (lipid profile, glycaemia, hepatorenal function).

Feature selection strategy

We employed a two-stage variable selection process integrating epidemiological principles with machine learning considerations. Initial univariable logistic regression identified candidate predictors associated with CVD incidence ($P < 0.05$). Qualifying variables then entered multivariable logistic regression with stepwise selection ($P < 0.05$) to identify independent predictors.

This approach served dual objectives. First, it yielded a parsimonious set of interpretable predictors with established clinical relevance. Second, it reduced dimensionality for machine learning models, minimizing overfitting risk while preserving statistically and biologically plausible features. Although tree-based algorithms inherently perform feature selection through recursive partitioning, pre-filtering enhanced computational efficiency and ensured clinical face validity.

We assessed collinearity for variables with correlation coefficients > 0.7 using variance inflation factors (VIF). When VIF exceeded 5, we retained variables with stronger clinical relevance or practical measurement feasibility. This consideration benefits both traditional regression—where multicollinearity inflates coefficient standard errors—and machine learning algorithms where correlated features may dilute variable importance estimates.

Eleven predictors were ultimately selected: geographical region, hypertension, dyslipidemia, hepatic disease, asthma, CESD-10 depression score, age, sleep duration, triglycerides, HDL cholesterol, and waist circumference.

Data pre-processing and quality control

We addressed missing data using multiple imputation by chained equations, generating five imputed datasets through 20 iterations each. Imputation methods were tailored to variable type: predictive mean matching for continuous measures, logistic regression for binary variables, and polytomous regression for the four-level geographical region variable. Following established guidelines, the outcome variable served as a predictor in the imputation models but was not itself imputed, thereby preserving the predictor-outcome associations in the original data. Trace plot inspection confirmed algorithm convergence across all chains. Statistical tests supported the missing-at-random assumption, with missingness patterns significantly linked to geographical region ($\chi^2 = 84.884$, $p < 0.001$) and urban-rural residence ($\chi^2 = 21.568$, $p < 0.001$), yet unrelated to cardiovascular outcome status ($\chi^2 = 2.993$, $p = 0.084$). We pooled estimates across imputed datasets according to Rubin's combination rules[15,30,31].

Continuous variables underwent Shapiro-Wilk normality assessment: normally distributed data were expressed as mean ± standard deviation with intergroup comparisons via t-tests; non-normally distributed data were reported as median (25th–75th percentile) with Mann-Whitney U or Kruskal-Wallis H tests. Categorical variables were presented as frequencies and percentages, with intergroup comparisons via χ^2 or Fisher's exact tests. Two-tailed tests were applied with significance threshold $\alpha = 0.05$.

Model construction and training

The cohort was partitioned into training ($n = 5{,}656$, 70%) and validation ($n = 2{,}424$, 30%) sets via stratified random sampling based on CVD occurrence status. Ten machine learning models were constructed in parallel: logistic regression (LR), support vector machine (SVM), gradient boosting machine (GBM), random forest (RF), extreme gradient boosting (XGBoost), light gradient boosting machine (LightGBM), categorical boosting (CatBoost), adaptive boosting (AdaBoost), neural network (NN), and k-nearest neighbours (KNN).

All models were trained on identical feature sets and data partitioning strategies. Hyperparameter optimisation employed 10-fold cross-validation with grid search ranges informed by literature review and exploratory analysis. For instance, random forest tree quantity (n_trees) was searched across 100–1000, with tree depth (max_depth) constrained to 5–20 layers; XGBoost learning rate ranged from 0.01 to 0.3.

Handling of class imbalance

Among the 5,657 participants in the training cohort, 1,246 (22%) experienced incident CVD during follow-up. While this represents a modest imbalance in outcome distribution, we elected not to employ synthetic resampling techniques (e.g., SMOTE) or algorithmic class weighting. This decision was guided by three considerations. First, the observed event rate approximates the true epidemiological burden of CVD in middle-aged and older Chinese adults, and preserving this natural prevalence is essential for generating well-calibrated absolute risk estimates suitable for clinical implementation. Second, artificial manipulation of class distributions—while potentially improving classification metrics—can introduce systematic miscalibration, particularly in the probabilistic outputs required for shared decision-making. Third, tree-based ensemble learners, which constituted the majority of candidate algorithms, exhibit documented resilience to moderate outcome imbalance through recursive partitioning mechanisms. Empirical validation of this approach was provided by calibration plots demonstrating close agreement between predicted and observed risks across probability strata, alongside decision curve analyses confirming net benefit over reasonable threshold ranges.

Model performance evaluation and selection

A multidimensional evaluation framework was employed: (1) Discrimination: area under the receiver operating characteristic curve (AUROC) with 95% confidence intervals, with inter-model comparisons via DeLong's test; (2) Classification metrics: accuracy, sensitivity, specificity, precision, and F1-score; (3) Calibration: calibration plots with Hosmer-Lemeshow χ^2 test for quantifying calibration error; (4) Clinical net benefit: decision curve analysis (DCA) assessing model performance relative to "treat all" and "treat none" strategies across risk threshold spectra.

All metrics were computed separately for training and validation sets. Model selection prioritised algorithms demonstrating maximal validation set AUROC, optimal calibration, and sustained positive net benefit across the 20%-80% risk threshold range. For models with comparable AUROCs (overlapping 95% confidence intervals), classification metrics were further compared. Statistical analyses utilised R version 4.2.0, with two-tailed significance threshold $P < 0.05$.

Statistical software

Analyses employed R 4.2.0. Core packages: mice (multiple imputation), catboost and lightgbm (ensemble learning), caret (unified ML framework), pROC and rms (discrimination and calibration), dcurves (decision curve analysis), shapviz v0.5.0 and kernelshap (SHAP interpretability), DynNom with Shiny (web calculator).

Model interpretability analysis

Interpretability analysis employed the SHapley Additive exPlanations (SHAP) framework. SHAP, grounded in game-theoretic Shapley value concepts, quantifies each feature's marginal contribution to predictions. Global SHAP value aggregation identified critical risk factors and their relative importance, whilst individual-level SHAP visualisation elucidated risk composition for specific patients, informing personalised intervention strategies.

Online risk assessment tool development

The validated optimal model was deployed as an interactive risk calculator using the Shiny framework. The front-end interface comprised 11 standardised input fields corresponding to core predictor variables in the final model. The back-end embedded the trained model (serialised in .rds format), executing data pre-processing, model prediction, and result visualisation. The system outputs individualised 9-year absolute CVD risk probability (0–100%). The web calculator runs on encrypted cloud infrastructure accessible across devices. We frame it as a screening facilitation and decision support tool that complements, rather than replaces, clinical judgment. Recalibration was deemed unnecessary: the validation sample—drawn from the same national sampling frame—exhibited adequate calibration without systematic bias requiring intercept adjustment. Users encounter an upfront interpretive statement emphasising that predicted probabilities capture statistical patterns in the training population and must be contextualised within broader clinical assessment.

Results

Baseline characteristics of study participants

The study enrolled 8,080 CHARLS participants completing 9-year follow-up, partitioned into training ($n = 5{,}657$, with 1,246 [22.0%] incident CVD cases) and validation ($n = 2{,}423$) sets at a 7:3 ratio. Supplementary Table S1 reports baseline distributions across 77 candidate predictor variables in the training set, alongside univariable and multivariable logistic regression results comparing CVD and non-CVD groups.

Demographic and geographical distribution characteristics

Gender composition differed between CVD and non-CVD groups, with female proportions of 55.9% and 51.8%, respectively. Whilst univariable analysis demonstrated lower CVD risk amongst males (OR = 0.85, 95% CI 0.75–0.96, $P = 0.009$), this association lost statistical significance following multivariable adjustment (adjusted OR [aOR] = 1.00, 95% CI 0.61–1.63, $P = 0.988$). Geographical distribution exhibited pronounced regional disparities: relative to northeastern China, eastern (aOR = 0.56, 95% CI 0.42–0.76, $P < 0.001$), central (aOR = 0.66, 95% CI 0.47–0.91, $P = 0.012$), and western regions (aOR = 0.54, 95% CI 0.40–0.74, $P < 0.001$) demonstrated significantly reduced CVD risk, suggesting elevated cardiovascular disease burden in northeastern provinces. Urban-rural differentials achieved statistical significance in univariable analysis but failed to reach significance threshold following multivariable adjustment (aOR = 0.93, $P = 0.389$).

Medical history

Hypertension history constituted the most potent predictor (aOR = 1.80, 95% CI 1.51–2.16, $P < 0.001$), with prevalence rates of 41.4% in the CVD group versus 17.5% in the non-CVD cohort. Dyslipidaemia demonstrated independent predictive value (aOR = 1.42, 95% CI 1.11–1.81, $P = 0.005$), affecting 14.8% of CVD cases compared with 5.5% of controls. Diabetes mellitus exhibited association with CVD in univariable analysis (OR = 2.38, $P < 0.001$), yet lost significance following multivariable adjustment (aOR = 0.94, $P = 0.718$). Hepatic disease history (aOR = 1.60, 95% CI 1.06–2.39, $P = 0.024$) and asthma history (aOR = 1.50, 95% CI 1.03–2.19, $P = 0.033$) both represented independent risk factors.

Metabolic and psychological factors

Depressive symptom scores exhibited positive correlation with CVD risk (aOR = 1.02, 95% CI 1.01–1.04, $P = 0.001$), with CESD-10 scores of 8.6 ± 6.2 in the CVD group versus 7.9 ± 6.1 in non-CVD participants. Metabolic syndrome demonstrated significant association in univariable analysis (OR = 3.98, $P < 0.001$), though

the effect attenuated following multivariable adjustment (aOR = 0.82, $P = 0.082$). All social engagement activity indicators failed to demonstrate significant associations.

Physiological measurements and laboratory parameters

Age constituted an independent predictor (aOR = 1.02, 95% CI 1.01–1.03, $P < 0.001$), with mean ages of 59.7 ± 9.1 years in the CVD group versus 58.2 ± 9.0 years in controls. Nocturnal sleep duration exhibited inverse correlation with CVD risk (aOR = 0.94, 95% CI 0.91–0.98, $P = 0.007$). Within the lipid profile, elevated triglyceride levels (191.2 ± 179.3 mg/dL in CVD versus 115.5 ± 65.6 mg/dL in non-CVD, aOR = 1.00, 95% CI 1.00-1.01, $P < 0.001$) and reduced HDL-cholesterol concentrations (44.9 ± 13.4 mg/dL in CVD versus 53.1 ± 15.4 mg/dL in non-CVD, aOR = 0.99, 95% CI 0.98–0.99, $P < 0.001$) were observed. Fasting glucose, glycated haemoglobin, uric acid, and C-reactive protein failed to achieve significance in multivariable models.

Waist circumference emerged as the most robust independent predictor (aOR = 1.05, 95% CI 1.04–1.07, $P < 0.001$), measuring 90.6 ± 10.1 cm in the CVD group versus 82.1 ± 11.9 cm in controls, with each 1-cm increment conferring 5% increased CVD risk. Systolic blood pressure, diastolic blood pressure, and body weight demonstrated no independent effects following multivariable adjustment.

Pain and functional status

Headache (OR = 1.43, 95% CI 1.19–1.71, $P < 0.001$), shoulder pain (OR = 1.39, 95% CI 1.16–1.67, $P < 0.001$), chest pain (OR = 1.45, 95% CI 1.10–1.91, $P = 0.009$), knee pain (OR = 1.42, 95% CI 1.19–1.70, $P < 0.001$), ankle pain (OR = 1.85, 95% CI 1.45–2.36, $P < 0.001$), and neck pain (OR = 1.56, 95% CI 1.23–1.97, $P < 0.001$) exhibited associations with CVD in univariable analyses, yet all lost significance following multivariable adjustment. Activities of daily living (ADL) and instrumental activities of daily living (IADL) scores showed no significant intergroup differences.

Final predictor variable determination

Based on univariable and multivariable logistic regression analyses (selection threshold $P < 0.05$), eleven core predictor variables were incorporated into machine learning models: geographical region, hypertension history, dyslipidaemia, hepatic disease history, asthma history, CESD-10 score, age, nocturnal sleep duration, serum triglycerides (TG), high-density lipoprotein cholesterol (HDL-C), and waist circumference. These variables encompassed multidimensional risk factors spanning geographical environment, medical history, psychological status, lifestyle behaviours, and biochemical metabolism.

Comparative discrimination performance of machine learning algorithms

Figure 2A,B present receiver operating characteristic (ROC) curves for ten algorithms across training and validation sets. In the validation set, random forest (RF) and gradient boosting machine (GBM) achieved the highest areas under the curve (AUCs) of 0.829 (95% CI 0.809–0.848) and 0.828 (95% CI 0.810–0.847), respectively, with extensively overlapping confidence intervals indicating comparable discrimination performance. LightGBM and CatBoost demonstrated validation AUCs of 0.818 and 0.810, respectively. Logistic regression exhibited training and validation AUCs of 0.780 and 0.792, respectively, with minimal generalisation loss. K-nearest neighbours (KNN) showed exceptional training performance (AUC = 0.959) but precipitously declined to 0.734 in validation, indicating pronounced overfitting.

Classification performance comparison

Table 1 presents classification metrics for each model. In the validation set, random forest attained the highest accuracy (0.770 (95% CI 0.655–0.797)), specificity (0.795 (95% CI 0.594–0.833)), precision (0.483 (95% CI 0.375–0.532)), and F1-score (0.565 (95% CI 0.518-0.600)), with sensitivity of (0.681 (95% CI 0.633–0.868)). GBM demonstrated maximal sensitivity (0.865 (95% CI 0.716–0.889)) but minimal specificity (0.620 (95% CI 0.602–0.774)), signifying misclassification of 38% of non-CVD individuals as high-risk. LightGBM exhibited balanced metrics, with accuracy, sensitivity, and specificity all approximating (0.728 (95% CI 0.564–0.816)).

Whilst Youden's index was employed to determine optimal classification thresholds for model comparison in this study, practical implementation requires context-specific threshold selection aligned with clinical objectives and resource constraints. The decision threshold should be calibrated based on: (1) relative costs of false-positive versus false-negative classifications in the target population; (2) available healthcare resources for downstream interventions; (3) patient risk tolerance and preferences; and (4) prevalence of CVD risk factors in the deployment setting. For primary prevention screening programmes with limited resources, higher thresholds (e.g., predicted risk > 30%) may be appropriate to prioritise high-risk individuals, thereby maximising specificity and positive predictive value. Conversely, comprehensive health assessment initiatives may adopt lower thresholds (e.g., > 15%) to enhance sensitivity and capture individuals who would benefit from early lifestyle modifications. The decision curve analysis (Fig. 3) provides quantitative guidance across the threshold spectrum, enabling stakeholders to select thresholds that optimise net benefit for their specific clinical context and target population characteristics.

Calibration assessment

Calibration plots (Fig. 2C,D) evaluated concordance between predicted probabilities and observed incidence rates. In the validation set, the random forest calibration curve most closely approximated the ideal diagonal, demonstrating high concordance between predicted probabilities and true risk. GBM exhibited systematic overestimation in high-risk intervals. Logistic regression displayed satisfactory calibration but inferior discrimination relative to tree-based models.

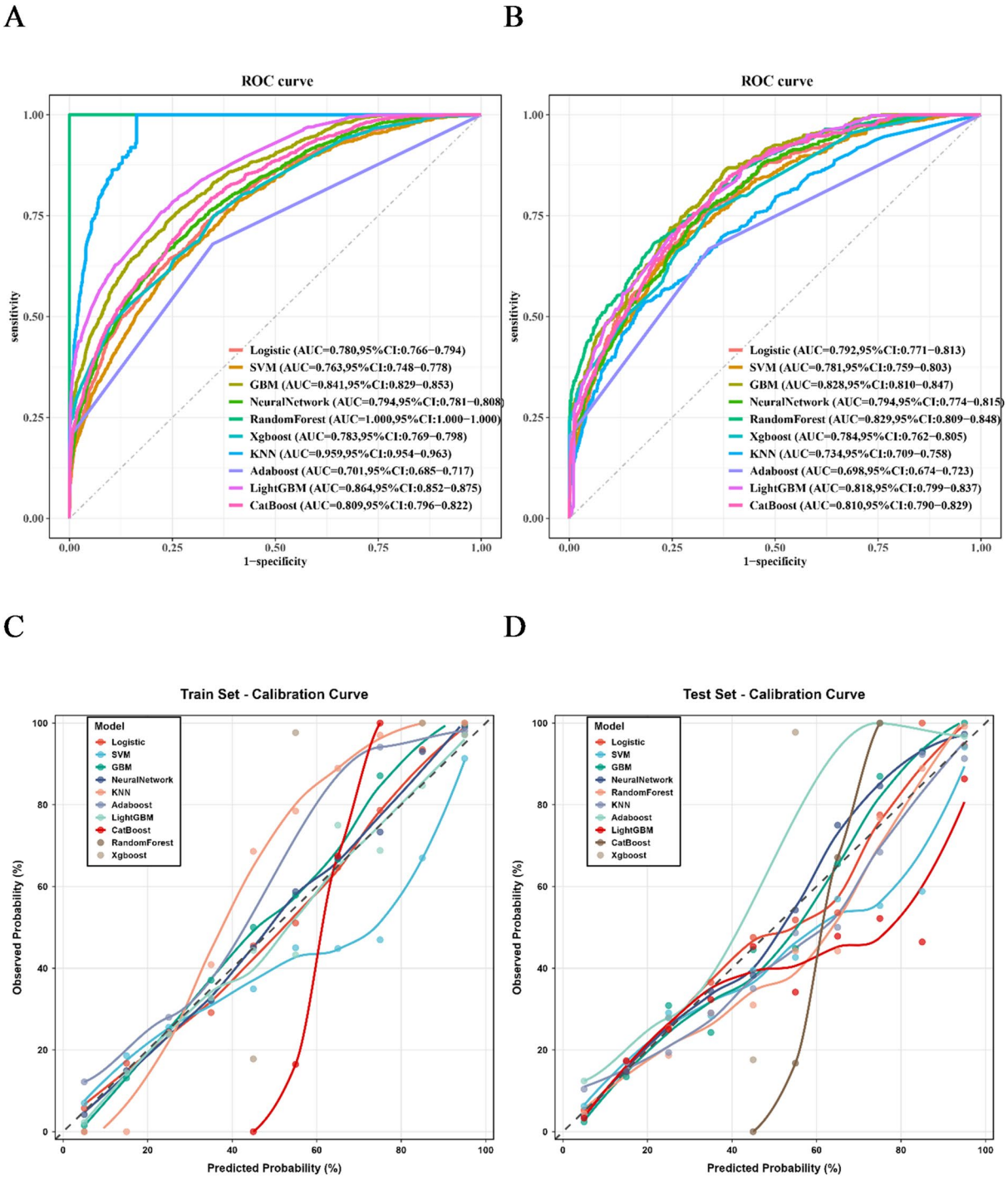


Fig. 2. Performance evaluation of machine learning models for predicting 9-year cardiovascular disease risk: receiver operating characteristic (ROC) curves and calibration plots in training and validation cohorts. Receiver operating characteristic (ROC) curves (**A**,**B**) illustrate the discrimination capacity of ten machine learning prediction models in the training set and independent validation set, respectively. Calibration plots (**C**,**D**) demonstrate concordance between predicted probabilities and observed event incidence rates, indicating satisfactory calibration across all risk strata.

Clinical net benefit analysis

Decision curve analysis (Fig. 3A,B) quantified clinical utility across risk threshold spectra. In the validation set, the random forest net benefit curve surpassed other models across the 10%-85% threshold range. Within the clinically relevant 20%-50% decision interval, random forest conferred 12%-18% incremental benefit compared with the "treat all" strategy.

Model	Accuracy (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	Precision (95% CI)	F1 (95% CI)	AUC (95% CI)
Logistic	0.718 (0.618–0.763)	0.711 (0.634–0.872)	0.720 (0.546–0.793)	0.417 (0.350–0.479)	0.526 (0.493–0.561)	0.792(0.771–0.813)
SVM	0.706 (0.627–0.769)	0.709 (0.605–0.842)	0.705 (0.567–0.808)	0.404 (0.348–0.490)	0.515 (0.486–0.557)	0.781(0.759–0.803)
GBM	0.674 (0.660–0.764)	0.865 (0.716–0.889)	0.620 (0.602–0.774)	0.391 (0.368–0.479)	0.539 (0.514–0.585)	0.828(0.810–0.847)
NeuralNetwork	0.697 (0.624–0.735)	0.758 (0.694–0.877)	0.680 (0.550–0.740)	0.400 (0.353–0.445)	0.524 (0.494–0.559)	0.794(0.774–0.815)
RandomForest	0.770 (0.655–0.797)	0.681 (0.633–0.868)	0.795 (0.594–0.833)	0.483 (0.375–0.532)	0.565 (0.518-0.600)	0.829(0.809–0.848)
Xgboost	0.683 (0.662–0.735)	0.752 (0.659–0.796)	0.663 (0.635–0.751)	0.387 (0.358–0.438)	0.511 (0.482–0.542)	0.784(0.762–0.805)
KNN	0.755 (0.654–0.778)	0.537 (0.488–0.723)	0.816 (0.634–0.849)	0.452 (0.350–0.505)	0.491 (0.455–0.528)	0.734(0.709–0.758)
Adaboost	0.662 (0.642–0.682)	0.668 (0.626–0.705)	0.660 (0.638–0.683)	0.357 (0.327–0.386)	0.465 (0.434–0.493)	0.698(0.674–0.723)
LightGBM	0.728 (0.631–0.785)	0.726 (0.634–0.890)	0.728 (0.564–0.816)	0.430 (0.354–0.512)	0.540 (0.501–0.579)	0.818(0.799–0.837)
CatBoost	0.673 (0.637–0.751)	0.831 (0.705–0.888)	0.628 (0.573–0.758)	0.387 (0.356–0.466)	0.528 (0.503–0.571)	0.816(0.790–0.829)

Table 1. Predictive performance of machine learning models for 9-year cardiovascular disease incidence: independent validation set results. Note: Performance metrics were calculated on the independent validation set ($n = 2,423$) comprising 30% of the total cohort. Optimal classification thresholds were determined using Youden's index from the training set. Random Forest achieved the highest accuracy (0.770) and specificity (0.795), while GBM demonstrated the highest sensitivity (0.865). All models were trained on 11 selected predictors derived from multivariable logistic regression (Table S1). CVD=Cardiovascular Disease; SVM=Support Vector Machine; GBM=Gradient Boosting Machine; KNN = K-Nearest Neighbors.

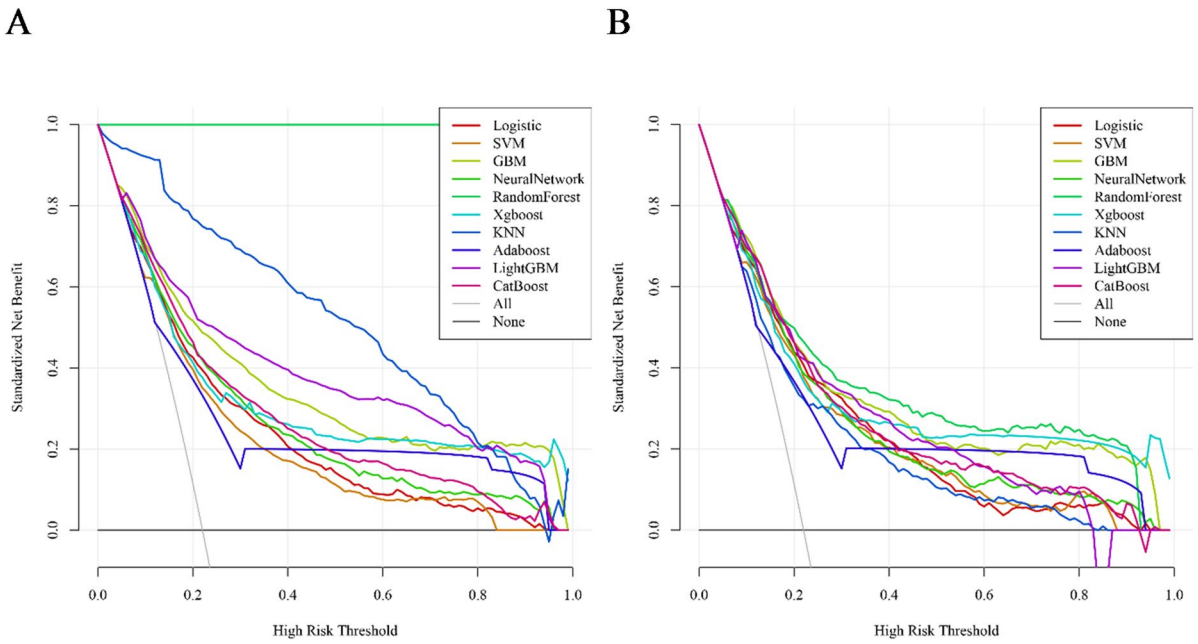


Fig. 3. Decision curve analysis (DCA) demonstrating clinical net benefit of machine learning models for 9-year cardiovascular disease (CVD) risk prediction. Net benefit is plotted against threshold probability in the training cohort (**A**) and validation cohort (**B**).

Optimal model selection

Integrating discrimination capacity, classification performance, calibration, and clinical net benefit, random forest was selected as the final prediction model based on the following rationale:

1. *Superior comprehensive classification performance* Whilst random forest demonstrated comparable discrimination to GBM, it exhibited significantly superior specificity (0.795 vs. 0.620), precision (0.483 vs. 0.379), and F1-score (0.565 vs. 0.526), effectively reducing false-positive rates.
2. *Optimal calibration* The random forest calibration curve displayed minimal deviation from the ideal diagonal, enabling direct utilisation of predicted probabilities for clinical risk communication, whereas GBM manifested systematic bias.
3. *Maximal clinical net benefit* Sustained delivery of highest net benefit across the 10%-85% decision threshold range, with pronounced advantages in moderate-to-high risk intervals (20%-50%).
4. *Reliable generalisation performance* Random forest controlled overfitting through bootstrap aggregating and random feature selection mechanisms, with validation performance (AUC = 0.829, 95% CI 0.809–0.848)

demonstrating superior discrimination capacity compared with logistic regression (AUC = 0.792, 95% CI 0.774–0.810), though the confidence intervals exhibit minimal overlap. The training AUC of 1.000 represents an inherent algorithmic characteristic and does not compromise generalisation capability.

Model interpretability analysis

The SHAP (SHapley Additive exPlanations) summary plot displays the magnitude and direction of each feature's contribution to model predictions. Features are ranked by mean absolute SHAP value, with higher values indicating greater importance. Each point represents an individual observation, with colour denoting feature value (red for high, blue for low) and horizontal position indicating impact on predicted risk. SHapley Additive exPlanations (SHAP) methodology was employed to elucidate the random forest model (Fig. 4A,B). The SHAP summary plot illustrated relative contributions of the eleven predictor variables. Waist circumference exhibited the highest mean absolute SHAP value, emerging as the most influential predictor and reflecting the pivotal role of abdominal adiposity in CVD risk assessment among middle-aged and older Chinese adults. Triglycerides, age, and hypertension history ranked sequentially thereafter, concordant with established cardiovascular epidemiological evidence and the multivariable regression findings of this investigation. HDL-C demonstrated inverse protective effects. Notably, CESD-10 score and nocturnal sleep duration demonstrated greater SHAP importance than certain established metabolic markers, indicating that psychobehavioural factors carry independent predictive signal. These rankings, however, reflect each variable's marginal contribution to model output rather than causal effect magnitudes—an interpretive distinction crucial to avoid overreading observational patterns as mechanistic pathways.

SHAP dependence plots (Fig. 5A,B) delineated relationships between individual variables and predicted risk. Waist circumference increment exhibited linear positive correlation with SHAP values, whilst elevated triglycerides and reduced HDL-C corresponded to risk augmentation and attenuation, respectively. The age-risk relationship displayed varying slopes across age strata. Nocturnal sleep duration manifested U-shaped associations, whereby both abbreviated and prolonged durations increased risk, suggesting protective threshold effects. These relational patterns furnish quantitative foundations for formulating individualised prevention strategies.

Web-based risk assessment tool

Leveraging the validated random forest model, a web-based CVD risk calculator was developed (Fig. 5C; accessible at https://zxy08887.shinyapps.io/make_web/). The tool requires input of eleven variables: demographic information (age, geographical region), medical history (hypertension, dyslipidaemia, hepatic disease, asthma), physical examination parameters (waist circumference), laboratory investigations (triglycerides, HDL-C), and lifestyle factors (nocturnal sleep duration, CESD-10 score). The system computes 9-year CVD incidence probabilities on demand. Three target use cases guide its design: first, screening in community and primary care settings to identify candidates for enhanced risk management; second, clinical decision support where predicted probabilities augment physician assessment without superseding it; third, informed self-evaluation by the general public, coupled with explicit recommendations for professional consultation before interpretation. A standardised preamble alerts all users that risk estimates reflect cohort-specific statistical patterns drawn from Chinese adults aged 45 + and do not constitute diagnostic conclusions.

Discussion

Leveraging 9-year longitudinal data from the nationally representative CHARLS cohort, this investigation systematically compared the performance of ten machine learning algorithms for cardiovascular disease risk prediction among middle-aged and older Chinese adults. The random forest model demonstrated optimal performance, achieving a validation AUC of 0.829, significantly exceeding that of conventional logistic regression

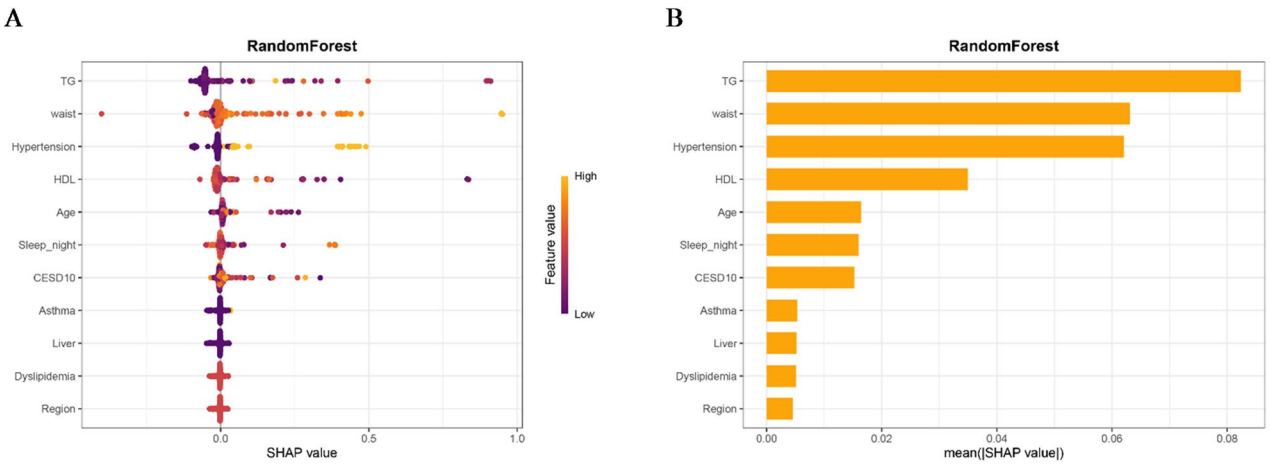


Fig. 4. SHAP global importance summary plot for the random forest model in predicting 9-year cardiovascular disease risk.

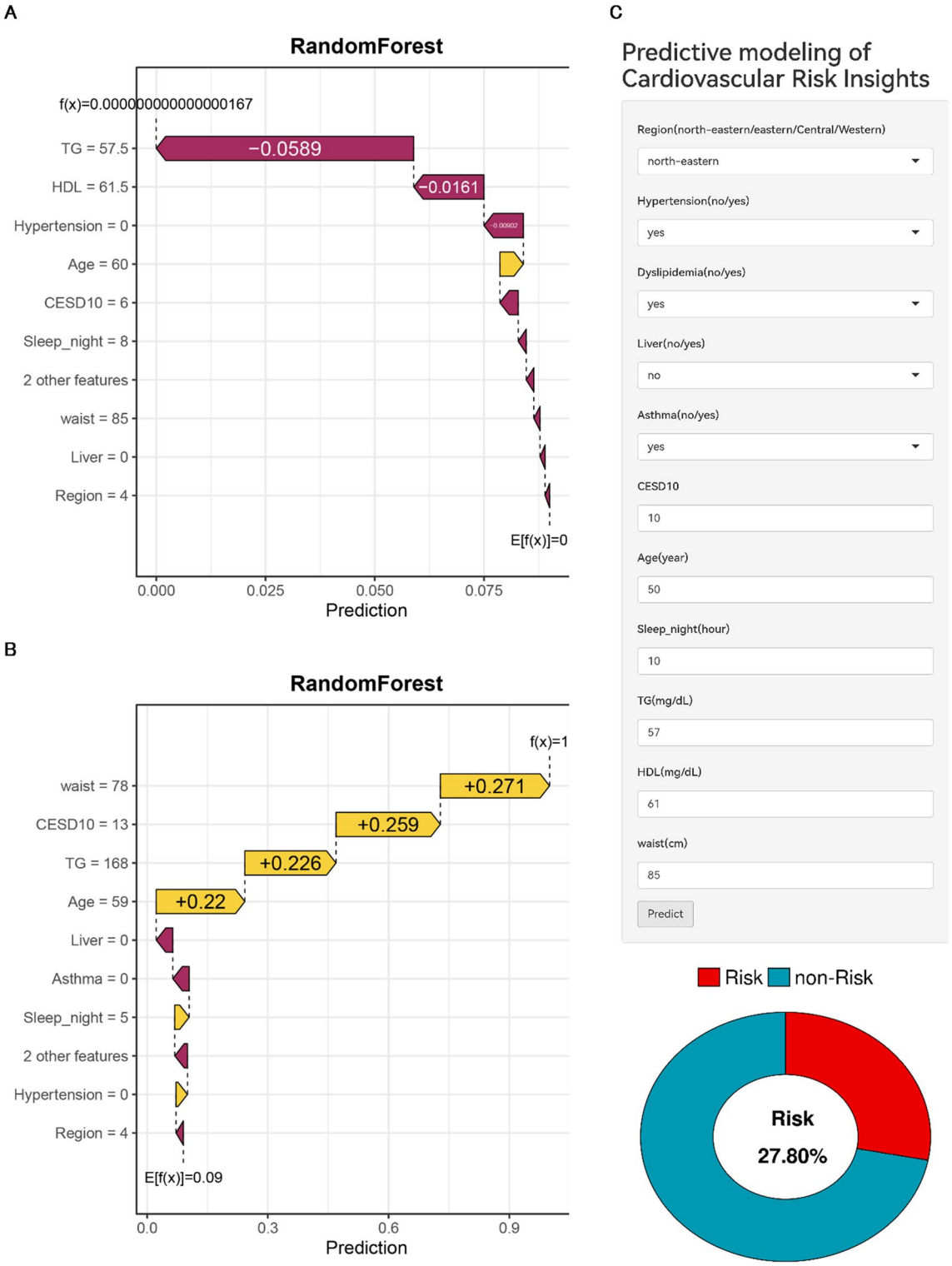


Fig. 5. SHAP dependence plots for key predictor variables and web-based risk calculator interface. (**A**,**B**) SHAP dependence plots illustrating the relationship between individual feature values and their impact on random forest model output for 9-year cardiovascular disease risk prediction. (**C**) Interactive interface of the web-based 9-year cardiovascular disease risk calculator powered by the random forest model, enabling real-time risk estimation through user input of clinical and lifestyle variables.

(AUC = 0.792). Through SHAP interpretability analysis, we identified waist circumference, triglycerides, age, and hypertension as the most influential predictors, whilst elucidating non-linear effect patterns and individualised contributions of each factor. The web-based risk assessment tool developed from this model enables instantaneous generation of individual 9-year CVD incidence probabilities, providing a pragmatic solution for community screening and clinical decision-making.

Comparison of model performance with existing literature

The predictive performance of our random forest model (AUC = 0.829) is comparable to or superior to recent domestic and international machine learning-based CVD prediction studies. Ambale-Venkatesh and colleagues[32] compared six machine learning algorithms for 10-year ASCVD risk prediction using the Multi-Ethnic Study of Atherosclerosis (MESA) cohort, with the optimal model (gradient boosting machine) yielding an AUC of 0.742, inferior to our findings. Dinh et al.[33] constructed a deep neural network model using UK Biobank data, achieving an AUC of 0.774 for 5-year CVD risk prediction. By contrast, our study maintained superior discriminative capacity despite a longer prediction horizon (9 years), suggesting model robustness. Notably, the deep learning model developed by Kakadiaris and colleagues[34] attained an AUC of 0.848 for 5-year CVD risk prediction; however, that investigation incorporated imaging parameters such as coronary artery calcium scoring, whereas our study achieved comparable performance utilising solely readily accessible clinical variables, rendering it more amenable to primary care settings.

Within Chinese population-based research, our results align closely with selected machine learning prediction models. Li et al.[35] employed XGBoost algorithms with data from over 100,000 participants in the Kailuan cohort to predict 10-year CVD risk, reporting a C-statistic of 0.812. Nevertheless, their predictor variables included laboratory indices such as uric acid and homocysteine, entailing higher data acquisition costs. Zhang and colleagues[36] utilised China Kadoorie Biobank (CKB) data to construct a random forest model predicting 10-year CVD risk with an AUC of 0.803, approximating our findings. However, the CKB cohort concentrated predominantly on specific geographical regions, whereas the national representativeness of CHARLS confers broader generalisability to our results. More importantly, most existing studies prioritise model performance comparisons whilst lacking in-depth elucidation of model decision-making processes, thereby constraining clinical acceptability.

Complementary evidence from Japanese population-based cohorts further contextualizes our findings within an East Asian framework. Using data from 7,389 participants in the Suita study, Vu and colleagues constructed stroke prediction models across five machine learning algorithms, with random forest again emerging as the top performer (AUC 0.73)[37]. SHAP decomposition identified age, systolic blood pressure, and estimated glomerular filtration rate as dominant predictors—a pattern broadly consistent with our own results despite differences in outcome definition. In a subsequent study of coronary heart disease prediction among 7,260 participants from Mitsuka City over 15 years of follow-up, the same group reported a random forest AUC of 0.73, with carotid intima-media thickness, blood pressure, and lipid profiles emerging as primary contributors, alongside novel markers such as serum calcium and body fat percentage[37,38]. The convergence of algorithm performance and core predictor hierarchies across these East Asian cohorts points to shared cardiovascular pathophysiology amenable to machine learning–based risk stratification. Nevertheless, population-specific predictors—carotid IMT in Japanese samples versus waist circumference and depressive symptoms in ours—highlight that predictor weights and risk thresholds require local calibration. These findings collectively argue against direct cross-population extrapolation of cardiovascular prediction models, even within ethnically proximate groups.

Our random forest model demonstrated superiority over the conventional China-PAR equation. China-PAR represents the most widely implemented CVD risk assessment tool domestically, with a C-statistic of 0.794 in the development cohort[10], marginally inferior to our findings. This discrepancy may arise from several considerations. Firstly, machine learning algorithms automatically capture non-linear relationships and higher-order interactions between variables, whereas traditional regression models necessitate a priori specification of these relationships by investigators[19]. Secondly, random forests effectively mitigate overfitting risks and enhance model generalisability through ensemble learning across multiple decision trees. Furthermore, our study incorporated psychobehavioural factors including depressive symptoms and sleep duration, extending beyond traditional risk factor paradigms. It merits emphasis that improvements in model performance require careful balance with interpretability and clinical practicability, which precisely underscores the significance of implementing SHAP analysis in this investigation.

Clinical significance of key predictors

SHAP analysis revealed waist circumference as the most influential predictor of 9-year CVD risk, a finding consonant with the central role of abdominal obesity in cardiovascular disease pathogenesis. As a convenient surrogate measure of visceral adipose tissue accumulation, waist circumference more accurately reflects metabolically active fat distribution compared with BMI[39]. Visceral adiposity precipitates insulin resistance, dyslipidaemia, endothelial dysfunction, and prothrombotic states through secretion of inflammatory cytokines (interleukin-6, tumour necrosis factor-α), adipokines (leptin, adiponectin), and free fatty acids, ultimately promoting atherogenesis[40]. Asian populations exhibit a propensity for greater visceral fat accumulation even at lower BMI levels, potentially accentuating the predictive value of waist circumference for CVD risk[41]. In our investigation, each 1-cm increment in waist circumference conferred a 5% increased CVD risk (adjusted OR = 1.05), with SHAP analysis demonstrating accelerated effect magnitude in higher value ranges, suggesting that even modest weight reduction might yield substantial cardiovascular benefits in individuals exceeding waist circumference thresholds. This observation substantiates the clinical relevance of Chinese adult waist circumference cut-points (≥ 90 cm for men, ≥ 85 cm for women) in CVD risk stratification[2].

The importance of triglycerides as the second most influential predictor was corroborated in this study. Epidemiological evidence demonstrates that hypertriglyceridaemia associates independently with CVD risk, potentially mediated through the atherogenic effects of remnant lipoproteins[42]. Mendelian randomisation studies support a causal relationship between triglycerides and CVD[43]. The Chinese Guidelines on Prevention and Treatment of Dyslipidaemia in Adults recommend borderline-high and elevated triglyceride cut-points of 1.7 mmol/L and 2.3 mmol/L, respectively[44]. Clinical practice should implement stratified intervention strategies according to patient triglyceride levels—lifestyle modifications for borderline elevation, with potential pharmacotherapy for marked elevation. Notably, HDL-cholesterol ranked lower in SHAP importance in our study, diverging from Western population research emphasising HDL as a protective factor, potentially reflecting distinctive lipid profile characteristics and genetic backgrounds in Chinese populations[45].

Hypertension and age, as classical risk factors, were reaffirmed in this investigation. Hypertension accelerates atherosclerotic progression through mechanical stress-induced endothelial injury, inflammatory pathway activation, and vascular remodelling promotion[46]. Hypertensive participants in our study demonstrated an 80% increased CVD risk (adjusted OR = 1.80), concordant with previous meta-analyses[47]. Age ranked as the fifth most important predictor in our study. The predictive value of age reflects not merely cumulative exposure duration to risk factors, but also biological processes including vascular senescence, endothelial dysfunction, and enhanced inflammatory states[48]. With advancing age, arterial stiffening accelerates, cardiovascular reserve capacity diminishes, and susceptibility to risk factors intensifies. These findings underscore the necessity for age-stratified risk assessment and comprehensive intervention strategies in middle-aged and elderly populations.

Independent contributions of psychobehavioural factors

This investigation identified depressive symptoms and sleep duration as possessing independent predictive value for CVD risk. Although psychobehavioural variables ranked modestly relative to traditional risk factors, their inclusion enhanced model discrimination, indicating clinically relevant predictive information beyond conventional metabolic and cardiovascular markers. Critically, SHAP-derived importance quantifies statistical contribution to predictions—not causal effect size[49]. The observed depression-CVD association aligns with epidemiological evidence documenting bidirectional relationships between the conditions. Hypothesised mechanisms linking depression to cardiovascular risk include HPA axis dysregulation with sustained cortisol elevation, sympathetic overdrive (elevated blood pressure and heart rate), systemic inflammation (CRP, IL-6), and prothrombotic states[50].Furthermore, depressed individuals frequently exhibit adverse lifestyle behaviours (smoking, sedentariness, poor dietary habits), further amplifying CVD risk[51]. Among middle-aged and older Chinese adults, depression recognition and treatment rates remain suboptimal[52]. Our findings suggest that incorporating convenient depression screening (such as the CES-D scale) into CVD risk assessment may facilitate identification of overlooked high-risk individuals.

Sleep duration emerged as an independent predictor of CVD risk in this study. Previous research indicates potential U-shaped associations between sleep duration and CVD risk. Short sleep duration (< 6 h) may increase CVD risk through sympathetic nervous system activation, inflammatory response induction, glucose-lipid metabolism disruption, and oxidative stress augmentation[53]. The mechanisms underlying elevated CVD risk associated with prolonged sleep (> 9 h) remain incompletely elucidated, potentially reflecting underlying health conditions (chronic diseases, depression) or socioeconomic disadvantage[54]. A meta-analysis encompassing 74 prospective studies demonstrated U-shaped associations between sleep duration and CVD risk, with optimal risk thresholds at 7–8 h[55]. Sleep duration demonstrated independent predictive value for CVD risk in this model. The observational design precludes distinguishing whether this reflects a causal effect, residual confounding, or reverse causation whereby early cardiovascular pathology disrupts sleep architecture. Given sleep's modifiability and ease of assessment, however, incorporating sleep history into risk conversations appears warranted despite mechanistic uncertainty.

Clinical value of SHAP interpretability

This investigation implemented SHAP technology to achieve transparent interpretation of machine learning models, bridging the chasm between predictive performance and clinical acceptability. Whilst traditional machine learning models demonstrate high predictive accuracy, their "black box" characteristics impede clinicians' comprehension of prediction logic, thereby constraining application[19]. The SHAP methodology, grounded in game-theoretic Shapley values, allocates marginal contributions to individual prediction outcomes for each predictor variable, possessing desirable mathematical properties including theoretical consistency and additivity[21]. Compared with conventional feature importance methods (such as Gini importance), SHAP not only provides global ranking but, more importantly, generates individualised interpretations—for any given patient, the model explicitly identifies which factors elevate risk, which exert protective effects, and the specific magnitude of each factor's contribution[56].

In clinical practice, such individualised interpretation holds substantial significance. For instance, for a patient with elevated waist circumference but normal blood pressure, SHAP force plots can intuitively demonstrate how waist circumference dominates the high-risk state, thereby guiding physicians to formulate targeted weight reduction programmes. Conversely, for elderly patients with well-controlled other factors, SHAP analysis may reveal that risk primarily originates from immutable age factors, thereby redirecting intervention priorities toward optimisation of other modifiable factors. This precise factor attribution enhances patients' comprehension and acceptance of risk assessment, strengthening self-management motivation. Within the shared decision-making framework, interpretable prediction results facilitate physician-patient communication, transforming patients from passive recipients into active participants in their own health management.

Variables with modest SHAP importance, such as asthma and geography, require careful interpretation. Their predictive signal reflects epidemiological associations detected in this cohort rather than validated biological

pathways. Clinicians and lay users alike should recognise these factors as contributing to probabilistic risk estimates—not as discrete causal levers amenable to intervention. Embedding this distinction into the web interface itself is critical to avoid misreading statistical outputs as mechanistic guidance.

Clinical application prospects and practical implications

The web-based risk assessment tool developed from the optimal model translates research findings into clinically pragmatic solutions. Requiring input of merely 11 readily accessible variables, the tool generates individual 9-year CVD incidence probabilities within seconds, applicable across diverse clinical scenarios. In community health centres or township health facilities, primary healthcare workers can utilise this tool to conduct cardiovascular risk screening, identify high-risk populations, and facilitate timely referrals to tertiary institutions. In general hospital outpatient settings, clinicians can integrate the tool into electronic medical record systems, achieving automated risk assessment and early warning to support clinical decision-making. In health management and medical examination institutions, the tool can provide personalised health consultation and lifestyle guidance for middle-aged and elderly individuals. Moreover, the public can access this tool via the internet for self-health assessment, enhancing cardiovascular disease prevention awareness.

The clinical value of this tool also manifests in cost-effectiveness considerations. Compared with traditional risk assessment instruments, our model does not depend on expensive laboratory tests (such as apolipoproteins, high-sensitivity C-reactive protein) or imaging examinations (such as coronary artery calcium scoring). Most variables can be obtained through medical history inquiry and simple physical examination, whilst lipid parameters constitute routine laboratory assessments. This low-cost characteristic confers excellent accessibility in China's vast primary care settings and economically underdeveloped regions. Decision curve analysis demonstrated that across risk threshold ranges of 10%-85%, the random forest model consistently outperformed "treat all" or "treat none" strategies, confirming clinical net benefit. Particularly under resource-constrained circumstances, this model can optimise resource allocation by concentrating limited intervention measures on genuinely high-risk populations, achieving precision prevention.

However, clinical application of the model necessitates cautious consideration. Firstly, risk prediction constitutes only one component of clinical decision-making and cannot completely supplant physicians' professional judgment and comprehensive assessment. Secondly, the model was developed in middle-aged and elderly populations; applicability to younger adults or special populations (such as pregnant women) requires further validation. Thirdly, the purpose of risk assessment is to initiate preventive interventions; subsequent follow-up management and patient adherence remain equally important. Finally, with evolving medical knowledge and changing risk factor profiles, the model requires periodic recalibration and updating to maintain predictive accuracy. Therefore, we recommend implementing this tool as a clinical decision support system rather than a replacement instrument in actual application, whilst establishing complementary quality control and effectiveness evaluation mechanisms.

Composite CVD endpoints trade specificity for power: while grouping coronary events, heart failure, and stroke increases incident cases for model training, it conflates conditions with partially distinct vascular mechanisms and risk factor hierarchies. Hypertension dominates stroke prediction; dyslipidemia drives coronary disease; heart failure reflects a spectrum from ischemic to non-ischemic etiologies. This heterogeneity may attenuate individual predictor effects and obscure the model's relevance to specific clinical presentations. Subtype-stratified analyses warrant future attention. For primary prevention—where treatment decisions target modifiable risk factors shared across outcomes—composite CVD prediction nonetheless remains appropriate and widely applied.

Principal strengths

The primary strength of this investigation resides in the representativeness of the data source. As a national cohort study, CHARLS encompasses community-dwelling populations across 28 provinces, providing a broad population foundation for model construction, partially offsetting the limitations of previous research predominantly confined to specific regions or healthcare institution patients. The 9-year follow-up period accumulated sufficient endpoint events, enabling the model to capture long-term CVD incidence risk, which possesses practical reference value for clinical decision-making.

Regarding predictor variable selection, the study integrated demographic characteristics, physiological indicators, biochemical parameters, and psychobehavioural factors, constructing a relatively comprehensive risk assessment framework that transcends traditional models' predominant reliance on biochemical indices. The model development process systematically compared ten machine learning algorithms, selecting the optimal model through cross-validation—this methodological design enhances result robustness.

Implementation of SHAP technology represents a salient feature of this investigation. This methodology achieves model decision process interpretability whilst maintaining machine learning's high predictive performance, partially alleviating concerns regarding "black box" models in clinical applications. SHAP analysis not only provides feature importance ranking but also generates individualised risk attribution; this transparency facilitates enhanced clinical acceptability. Furthermore, the web-based risk assessment tool developed in this study translates research findings into practical products, providing convenient pathways for clinical application. The study employed rigorous statistical methods in data processing (multiple imputation) and model evaluation (AUROC, calibration curves, decision curves), demonstrating satisfactory methodological quality.

Limitations

Several methodological considerations merit acknowledgement. Our reliance on self-reported physician diagnoses for CVD ascertainment introduces potential misclassification despite CHARLS quality assurance protocols. Linkage with administrative health databases or mortality registries would strengthen endpoint

validation in future iterations. The CHARLS sampling architecture—probabilistic recruitment of community-dwelling adults aged ≥ 45 years—systematically underrepresents younger populations, institutionalized individuals, and geographically concentrated high-risk cohorts. Stringent eligibility criteria requiring complete 9-year follow-up and comprehensive baseline biochemistry further constrained our analytical sample; participants lost to follow-up, frequently those experiencing adverse health trajectories or facing healthcare access barriers, were necessarily excluded. Such selection may attenuate observed risk gradients.

Predictor availability was circumscribed by survey instrumentation, precluding integration of genetic architecture (polygenic scores) or emerging biomarkers (high-sensitivity troponin, NT-proBNP). While these additions might enhance discrimination, our emphasis on routine clinical measures aligns with pragmatic implementation priorities. Biennial assessment cycles permitted only baseline modelling; risk factor trajectories—blood pressure fluctuations, lipid responses to therapy, weight dynamics—remain uncharted. Time-varying covariate models could capture such evolution but substantially compound analytical and implementation complexity.

The decade separating baseline data collection (2011–2012) from contemporary practice has witnessed considerable epidemiological shifts: evolving risk factor distributions, recalibrated diagnostic criteria, and expanding therapeutic landscapes. Regional variations in healthcare infrastructure, socioeconomic gradients, and dietary patterns transcend our geographical categorization, introducing unmeasured heterogeneity that likely varies across deployment settings. External validation in independent cohorts remains outstanding; planned collaboration with complementary Chinese platforms (CKB, CHARLS sister studies) should illuminate generalizability boundaries. Populations demonstrating substantially different baseline profiles or CVD incidence necessitate local recalibration—preserving the predictor hierarchy while adjusting intercepts, as discrimination transportability typically exceeds that of absolute risk estimates. Our web platform accommodates such coefficient refinement. Prospective performance monitoring should trigger recalibration upon detecting meaningful calibration degradation.

Random forest implementation demands greater computational infrastructure than traditional approaches; resource-constrained primary care settings may find logistic regression-based scoring systems more tractable. Finally, as an observational investigation, reported associations cannot establish causality. Bidirectional relationships warrant consideration—depression and sleep disturbances may represent prodromal CVD manifestations rather than purely etiological factors—and unmeasured confounding persists despite multivariable adjustment.

Future research directions

Based on these findings, several directions merit further exploration. Independent external cohort validation represents a critical step in assessing model generalisability. Model predictive performance should be evaluated across different regions, urban-rural settings, ethnicities, and socioeconomic populations, with necessary model recalibration according to validation results.

Dynamic risk prediction constitutes an important avenue for enhancing model accuracy. Utilising repeated measurement data from multiple follow-up waves (longitudinal changes in blood pressure, lipids) to update individual risk assessments can achieve "dynamic monitoring and timely warning". Statistical methods such as joint models or Landmark models can construct such dynamic prediction frameworks, though their implementation demands higher technical requirements.

Integration of multi-omics data (genomics, proteomics, metabolomics) and emerging technologies (wearable devices, medical imaging artificial intelligence) holds promise for constructing multimodal fusion precision prediction models. With precision medicine advances, multilevel biological information integration may achieve more refined risk stratification. However, balance must be sought among data acquisition costs, technical complexity, and clinical accessibility.

Prospective intervention trials represent a necessary component for validating model clinical value. Prediction models established through observational research require verification through randomised controlled trials: whether risk-stratified intervention strategies based on models truly reduce CVD incidence and mortality, and their cost-effectiveness ratios. Only through intervention trials can final confirmation be obtained regarding whether model-guided clinical decisions improve patient outcomes.

Intelligent upgrading of clinical decision support systems represents an approach to enhancing model practicability. Deep integration of risk prediction models with electronic medical record systems, combined with natural language processing technology to automatically extract predictor variables from medical record text, can achieve automated risk assessment and personalised intervention recommendation, alleviating healthcare worker burden.

Model fairness and ethical considerations warrant attention. Assessment is needed regarding whether models exhibit systematic prediction biases across different population subgroups, avoiding algorithmic exacerbation of health inequities. Particularly for ethnic minorities and low socioeconomic status populations, model applicability and fairness must be ensured. Other explainable artificial intelligence methods (such as LIME, attention mechanisms) merit exploration. Through comparison of different interpretation method merits, standardised frameworks for interpretability assessment can be established, which holds substantial significance for promoting responsible artificial intelligence application in the medical domain.

Conclusions

In summary, leveraging long-term follow-up data from the nationally representative CHARLS cohort, this investigation successfully constructed a long-term CVD risk prediction model applicable to middle-aged and elderly Chinese populations. The random forest algorithm demonstrated optimal performance, with predictive capacity significantly exceeding traditional statistical methods. Through SHAP explainable artificial intelligence

technology, we not only quantified the relative importance of predictor factors but also elucidated non-linear effect patterns and individualised contributions, bridging the contradiction between machine learning's "high performance" and "black box" characteristics. Waist circumference, triglycerides, age, and hypertension emerged as the most critical predictors, whilst independent contributions of psychobehavioural factors including depressive symptoms and sleep duration expanded the dimensions of CVD risk assessment. The web-based risk assessment tool developed in this study possesses characteristics of convenience, rapidity, and low cost, applicable across diverse scenarios including community screening, clinical decision-making, and self-health management, providing scientific evidence and pragmatic solutions for precision prevention of cardiovascular disease in China's middle-aged and elderly populations. Future work necessitates further validation of model external validity in independent cohorts and assessment of actual impact on clinical outcomes through prospective intervention trials.

Data availability

The datasets analysed during this study are publicly available from the CHARLS repository ([http://charls.pku.edu.cn/en](http:/charls.pku.edu.cn/en)). Access to restricted data components may be obtained through formal application to the CHARLS team.

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Author contributions

XYZ and GLY conceptualized the study design and developed the analytical framework. XYZ, WL, and XYP performed the data preprocessing, feature engineering, and statistical analyses. XYP implemented the machine learning algorithms and SHAP interpretability analysis. TL and LTY validated the model performance and conducted the comparative analysis of different algorithms. XYZ drafted the initial manuscript with contributions from WL and LTY. GLY provided critical revision of the methodology, secured research resources, and supervised the overall research direction. All authors reviewed, edited, and approved the final manuscript for publication.

Declarations

Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

The CHARLS baseline and all follow-up surveys received ethical approval from the Biomedical Ethics Review Committee of Peking University (IRB00001052-11015). Written informed consent was obtained from all participants prior to enrolment.

Additional information

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